

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 35%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.	BL2 – Understand	5
2	Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.	BL2 – Understand	5
3	Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.	BL2 – Understand	5
4	Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.	BL1 – Remember	5
5	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand	5
6	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand	5
7	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand	5
8	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and	BL1 – Remember	5

	transaction manager.		
9	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand	5
10	Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.	BL2 – Understand	5

Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness
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Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides

Introduction to DBMS

Definition

A Database Management System (DBMS) is software that enables users to define, create, maintain, and control access to a database. It acts as an intermediary between the user (or application program) and the physical database, allowing data to be stored, retrieved, and manipulated efficiently and securely.



The DBMS interprets requests from the user, retrieves or updates the required data in the database, and returns the result — shielding the user from the complexity of physical storage.

Objectives of a DBMS

1

Minimize Data Redundancy

Centralize data storage so the same fact is not duplicated across multiple files.

2

Ensure Data Integrity

Enforce accuracy and consistency through constraints, validation rules, and referential integrity.

3

Provide Concurrent Access

Allow multiple users to access and update data simultaneously without conflicts, using concurrency control.

4

Support Data Independence

Let applications remain unaffected by changes to how data is physically or logically stored.

Advantages of a DBMS

- Reduced data duplication and improved data sharing across applications
- Enforced integrity constraints that keep data accurate and reliable
- Stronger security through authentication and role-based access control
- Reliable backup and recovery mechanisms to protect against data loss
- Efficient concurrent access allowing many users to work simultaneously
- Standardized data access using query languages such as SQL

Real-World Applications

Banking

Account management, transactions, and real-time fraud detection

Healthcare

Electronic health records shared across hospital departments

E-Commerce

Product catalogs, inventory tracking, and order management

Airlines

Reservation systems, ticketing, and scheduling

Education

Student information systems and academic records

Telecom

Subscriber data, billing, and call records management

Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.

Answer:

Aspect	File-Based System	DBMS
Data Redundancy	High – same data duplicated across files	Minimized – centralized data storage
Data Integrity	Difficult to enforce, error-prone	Enforced via constraints and rules
Concurrent Access	Not well supported, risk of conflicts	Supported through concurrency control
Security	Limited, file-level protection only	Robust, role-based access control
Backup & Recovery	Manual and unreliable	Automated and transaction-safe

Case Study – Banking:

Traditional bank branches once maintained separate ledger files for each branch, causing duplicate customer records and delayed reconciliation. Modern banks use centralized DBMS (e.g., Core Banking Systems) so that a customer's balance updates instantly across all branches and ATMs, ensuring consistency and real-time fraud detection.

Case Study – Healthcare:

Hospitals earlier stored patient records in separate paper/file systems per department (pharmacy, lab, billing), causing mismatched patient histories. Electronic Health Record (EHR) systems built on DBMS integrate all departments, allowing doctors instant access to accurate, unified patient data, improving diagnosis and reducing medical errors.

Question 3

BL2 – Understand | Marks: 5

Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.

Answer:

External Level (View Level):

Closest to the user; defines how individual users see their portion of data. E.g., a bank teller's screen shows only account balance and transactions, not loan details.

Conceptual Level (Logical Level):

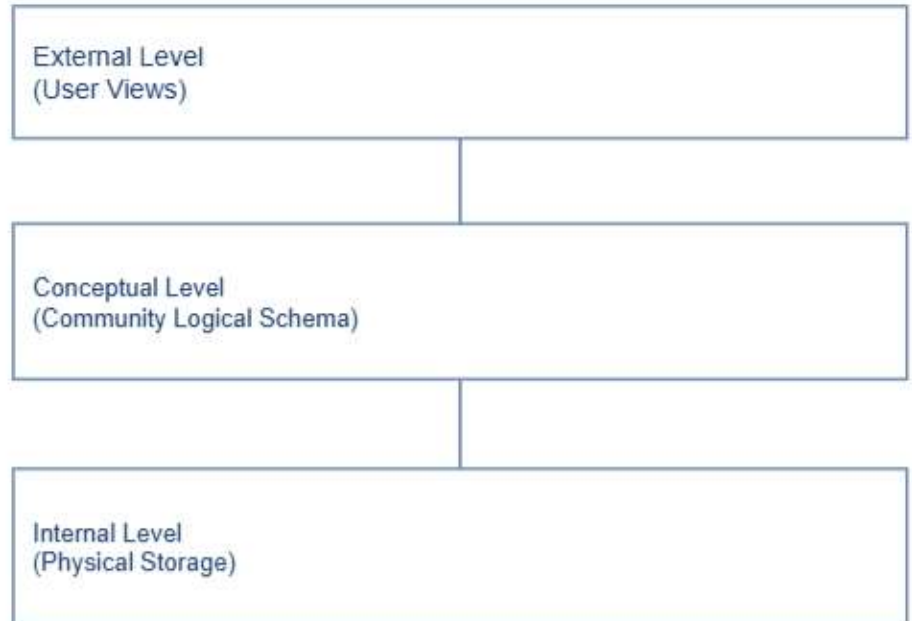
Describes the structure of the whole database for the community of users — entities, relationships, and constraints — independent of physical storage.

Internal Level (Physical Level):

Describes how data is actually stored — file structures, indexing, and access paths on disk.

Practical Example:

In an online library system, a student's external view shows only book search results; the conceptual level defines Book, Member, and Loan relationships; the internal level manages B-tree indexes on Book_ID for fast retrieval.



Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.

Answer:

Hierarchical Model:

Organizes data in a tree-like parent-child structure; each child has exactly one parent. Used in early IBM IMS systems.

Industry use: File systems, XML document structures, telecom subscriber hierarchies.

Network Model:

An extension of the hierarchical model where a child (member) can have multiple parents (owners), represented as a graph.

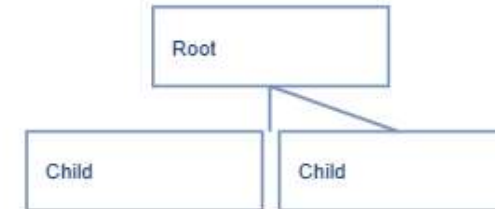
Industry use: Airline reservation systems (CODASYL-based legacy systems).

Relational Model:

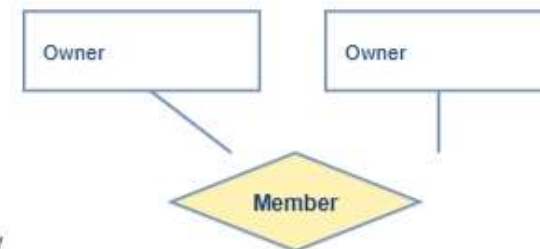
Data is organized into tables (relations) of rows and columns; relationships are managed using keys (primary/foreign). Most widely used today.

Industry use: MySQL, Oracle, PostgreSQL — banking, e-commerce, and ERP systems.

Hierarchical



Network



Relational

ID	Name
101	Asha
102	Ravi

Question 5

BL2 – Understand | Marks: 5

Create and present an ER diagram for an Online Library System, explaining each component clearly.

Answer:

Entities:

Member, Book, Author, Loan.

Member (Member_ID, Name, Email, Phone) — registered users of the system.

Book (Book_ID, Title, ISBN, Category, Copies_Available) — catalog items.

Author (Author_ID, Name) — writes one or more books.

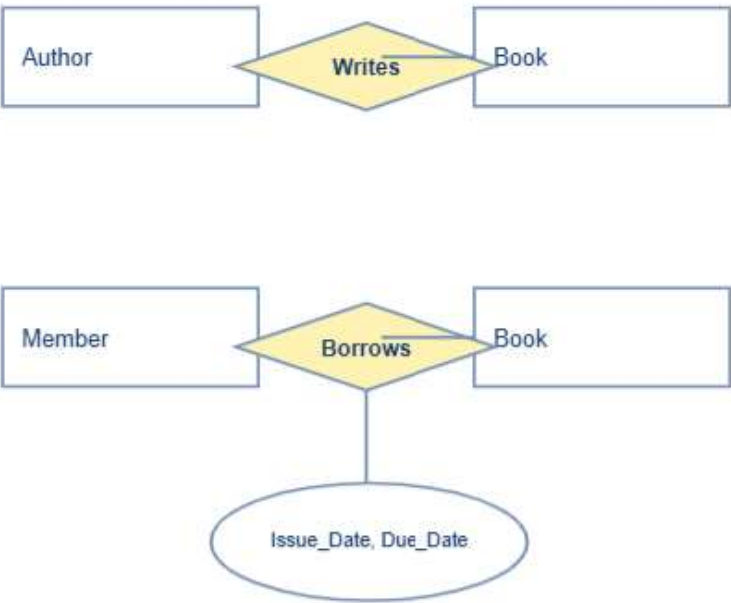
Relationships:

Writes: Author–Book (Many-to-Many) — a book can have multiple authors.

Borrows: Member–Book (Many-to-Many via Loan) with attributes Issue_Date and Due_Date.

Keys:

Member_ID and Book_ID are primary keys; Loan uses a composite key (Member_ID, Book_ID, Issue_Date).



Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.

Answer:

Specialization:

Top-down process of defining subclasses from a superclass based on distinguishing characteristics. E.g., the Person entity is specialized into Student and Employee.

Generalization:

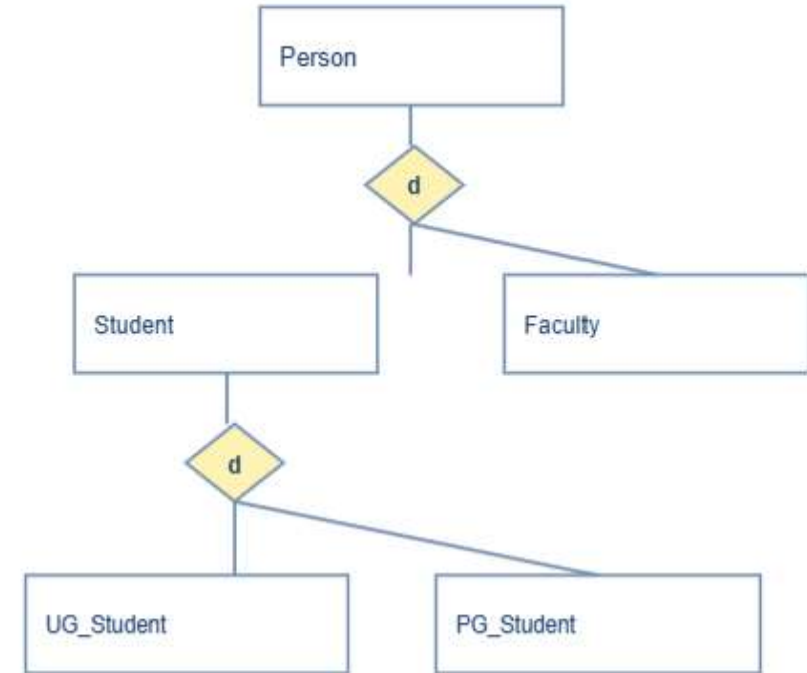
Bottom-up process where common attributes of Student and Faculty (e.g., ID, Name, Address) are combined into a higher-level entity, Person.

University Case Study:

Superclass Person (Person_ID, Name, Address) generalizes Student (Roll_No, Program) and Faculty (Emp_ID, Department).

Student is further specialized into UG_Student and PG_Student (disjoint, total specialization).

This models real-world overlap and disjoint constraints efficiently, avoiding redundant attributes.



d = disjoint, total specialization

Present the concept of data independence (logical and physical) and its significance in DBMS architecture.

Answer:

Data Independence:

The capacity to change the schema at one level of the database without having to change the schema at the next higher level. It maps directly onto the ANSI/SPARC three-level architecture.

Logical Data Independence:

Ability to modify the conceptual schema (e.g., adding a new entity or attribute) without altering external schemas or application programs. Harder to achieve since applications depend closely on the logical structure.

Physical Data Independence:

Ability to change the internal schema (e.g., storage structure, indexing method, or moving to new hardware) without affecting the conceptual or external schema. Easier to achieve.

Significance:

- Reduces application maintenance cost when the database structure evolves.
- Allows database administrators to optimize storage/performance without disrupting users.
- Supports scalability and long-term flexibility of enterprise systems.

Question 8

BL1 – Remember | Marks: 5

Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.

Answer:

Query Processor:

Interprets and translates user queries (e.g., SQL) into low-level instructions the database engine can execute. Includes the query parser, optimizer (choosing the most efficient execution plan), and evaluation engine.

Storage Manager:

Acts as an interface between the data stored in the database and the queries received. Responsible for the file manager (allocates disk space), buffer manager (manages data in memory cache), and authorization/integrity manager (enforces constraints and access rights).

Transaction Manager:

Ensures that the database remains in a consistent state despite system failures and concurrent access. Enforces ACID properties (Atomicity, Consistency, Isolation, Durability) using concurrency control and recovery management subsystems.

Together, these components allow a DBMS to efficiently process user requests while maintaining data accuracy, security, and reliability.



Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.

Answer:

Aspect	Strong Entity	Weak Entity
Primary Key	Has its own primary key	No primary key of its own; relies on identifying owner
Existence	Exists independently in the database	Cannot exist without its owner (strong) entity
Representation	Single rectangle in ER diagram	Double-outlined rectangle in ER diagram
Relationship	Related to weak entities via identifying relationship	Connected using a double diamond identifying relationship
Example	Customer (Customer_ID) in a bank database	Dependent (Dependent_Name) that exists only via Employee (Emp_ID)

Real-World Example: In an Employee database, Employee is a strong entity identified uniquely by Emp_ID. Dependent is a weak entity — a dependent's record (e.g., Dependent_Name, Relationship) only makes sense in the context of a specific Employee, and the composite key (Emp_ID, Dependent_Name) uniquely identifies each dependent record.

Question 10

BL2 – Understand | Marks: 5

Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.

Answer:

Core Entities:

Store, Product, Employee, Supplier, Sale.

Store (Store_ID, Location, Manager) — branch outlets of the chain.

Product (Product_ID, Name, Price, Category) — items sold across stores.

Employee superclass specialized into Cashier and Manager (disjoint, total specialization) — reflects distinct roles.

Relationships:

Supplies: Supplier–Product (Many-to-Many) with attribute Supply_Date.

Sells: Store–Product via Sale (Many-to-Many) capturing Quantity and Sale_Date.

Works_At: Employee–Store (Many-to-One) — each employee belongs to one store.

Design Walkthrough Outcome:

The EER model captures role hierarchies (Employee specialization) and many-to-many transactions (Sales, Supplies), giving a scalable schema for multi-branch retail operations with centralized reporting.

